Fiscal Year:	FY 2021	**Task Last Updated:**	FY 04/11/2021
PI Name:	Narayanan, Ranga Ph.D.		
Project Title:	A Novel Way to Measure Interfacial Tension Using the Electrostatic Levitation Furnace (ELF)		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	ranga@ufl.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	352-281-5382
Organization Name:	University of Florida		
PI Address 1:	Chemical Engineering, 205, PST		
PI Address 2:			
PI Web Page:			
City:	Gainesville	**State:**	FL
Zip Code:	32611	**Congressional District:**	3
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2015 NNH15ZTT002N MaterialsLab Open Science Campaigns for Experiments on the International Space Station
Start Date:	05/22/2017	**End Date:**	05/21/2022
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Sansoucie, Michael	**Contact Phone:**	256.544.5269
Contact Email:	michael.p.sansoucie@nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX17AL27G		
Performance Goal No.:			
Performance Goal Text:			
	A novel method using an Electrostatic Levitator (EL) is proposed to determine the surface tension of liquid/gas systems. This key thermophysical property is key to predicting optimum conditions for additive manufacturing, materials processing, and welding operations in space and on ground. The novel method for measurement of surface tension proposed involves the use of electrostatically forced Faraday instability. In this method, an oscillatory electrostatic forcing is applied to a liquid system, resulting in sudden pattern formation at the interface, i.e., an instability is caused. The "instability" is a result of resonance between the imposed forcing frequency and the natural frequency of the system. The critical amplitude and mode shapes are predictable from theory and compared with experiments. The measurement of surface tension directly aligns with the goals of the Thermophysical Property Measurement emphasis of the current NRA (NASA Research Announcement).		

Task Description:	Faraday experiments by this team using mechanical forcing have shown excellent agreement between theory and experimental results. However, in the case of mechanical forcing, the force of gravity overwhelms the effects of surface tension on the system stability and therefore it is almost impossible to determine this property with accuracy. Electrostatic forcing counters gravity, allowing the effects of surface tension to play a significant role in the ultimate determination of the system stability. Accordingly, experimentally mapped stability curves can be fit accurately with theoretical curves, allowing for the determination of the surface tension with great accuracy. The proposing team has successfully used electrostatic forcing in ground equipment to show that Faraday waves can be obtained as per theoretical predictions. The current state of the art uses an open container geometry to hold the liquid. However, a levitated sphere may also be used, and our proposal uses this containerless method. We therefore propose experiments with an important novel aspect. We propose to use ground-based facilities at JAXA (Japan Aerospace Exploration Agency) and NASA Marshall Space Flight Center (MSFC) to obtain tension for liquids using electrostatic Faraday forcing in using spherical drops by oscillating the drop over a large range of frequencies until the drop resonates with different modal patterns. The surface tension obtained from resonance at one modal patterns will be verified from deduced surface tension from another modal patterns. This will provide a self-consistent bench mark for determining surface tension. In summary, the research allows an alternate new method, viz., Faraday instability, for the determination of surface tension. Using electrostatic forcing instability will permit an independent confirmation of measurements obtained by the single-pulse oscillating-drop technique where decay rates are measured and related to surface tension and viscosity. The proposed method will also provide a slef consistent bench mark. Ground-based research using the electrostatic levitator available at the NASA MSFC and JAXA, and space experiments on ELF in the International Space Station (ISS) will provide accurate data. In addition to being an alternative method for surface tension measurement, the proposed technique also has versatility as a potential means in measuring surface tension for high viscosity materials and two-liquid interfacial tension on ground as well as in space. The proposing team has recently collaborated successfully with JAXA and has conducted experiments to study the feasibility of electrostatic Faraday instability. These results form the basis for the current proposal.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Accurate surface tension values lead to transformative impact with space and Earth benefits such as a) Additive manufacturing of metals, where knowledge of surface tension is pivotal for predictive modeling to avoid rippling during solidification and possible wrinkling and shrinkage.. Human Exploration: Technology Area (TA) TA7.2.3.5: Welding, TA12.1.2: Computationally-Designed Materials, TA12.4.2.5: Additive Manufacturing. b) Space enabling operations. One example is joining operations such as soldering, where the surface plays an important role in the structure and stability of the weld. Accurate values for the surface tension will enhance the safety and effectiveness of this type of operation in space where the microstructures formed are different than those on Earth where gravity is present. c) Semiconductor crystal growth, where the interfacial tension between a liquid melt and liquid encapsulant plays a large role in determining conditions where convection is present.
Task Progress:	A self consistent benchmark to the measurement of surface tension of high temperature liquid metals was developed with a proof-of-concept. In this method we have used the idea of resonance of levitated drops to create various patterns or modes. Each mode is correlated to surface tension. When the parametric frequency equals the natural frequency, the system undergoes an instability that is manifested by a sudden change of its configuration. The natural frequency is related to the surface tension and so the onset of the instability is used to determine the surface tension. By sweeping through a range of frequencies we can obtain different modes and each mode uniquely determines the surface tension. We can thus get a self consistent way of determining surface tension. This technique was demonstrated for several liquid levitated drops at JAXA and at MSFC. A paper was published and a patent disclosure filed.
Bibliography Type:	Description: (Last Updated: 06/17/2026)
Abstracts for Journals and Proceedings	Dinesh B, Brosius N, Narayanan R. "Nonlinear Instability of Resonance in Thin Films with Patterned Walls: Theory and Experiments." Presented at the 2020 American Institute of Chemical Engineers Annual Meeting, Virtual, November 16-20, 2020. AIChE Abstract Book, 2020 American Institute of Chemical Engineers Annual Meeting, Virtual, November 16-20, 2020. , Nov-2020
Abstracts for Journals and Proceedings	Dinesh B, Narayanan R. "Pattern Formation in Electrostatic Instability: Weakly Nonlinear Analysis." Presented at the 2020 American Institute of Chemical Engineers Annual Meeting, Virtual, November 16-20, 2020. AIChe Abstract Book. 2020 American Institute of Chemical Engineers Annual Meeting, Virtual, November 16-20, 2020. , Nov-2020
Abstracts for Journals and Proceedings	Dinesh B, Narayanan R. "Electrostatic interfacial instability: Weakly nonlinear analysis." Presented at the 73rd Annual Meeting of the APS Division of Fluid Dynamics, Virtual, November 22–24, 2020. Bulletin of the American Physical Society. 2020;65:Abstract: G02.00008. http://meetings.aps.org/Meeting/DFD20/Session/G02.8 ; accessed 4/13/21. , Nov-2020
Abstracts for Journals and Proceedings	Dinesh B, Narayanan R. "Weak Nonlinear Analysis of the Rayleigh-Taylor Instability in Linear Viscoelastic Fluids." Presented at the 73rd Annual Meeting of the APS Division of Fluid Dynamics, Virtual, November 22–24, 2020. Bulletin of the American Physical Society. 2020;65:Abstract: Z08.00001. http://meetings.aps.org/Meeting/DFD20/Session/Z08.1 ; accessed 4/13/21. , Nov-2020

Abstracts for Journals and Proceedings	Dinesh B, Narayanan R. "The Rayleigh-Taylor instability revisited with surprising and unintuitive results." Presented at the Complex Fluids 2020, Indian Institute of Technology (IIT) Bombay, Mumbai, India, Virtual, December 10-12, 2020. Book of Abstracts, COMPFLU 10. Complex Fluids 2020, Indian Institute of Technology (IIT) Bombay, Mumbai, India, Virtual, December 10-12, 2020. , Dec-2020
Abstracts for Journals and Proceedings	Brosius N, Ward K, Wilson E, Karpinsky Z, SanSoucie M, Ishikawa T, Matsumoto S, Narayanan R. "Surface Tension Measurement Using Resonance of Levitated Molten Droplets." Presented at the 2020 American Institute of Chemical Engineers Annual Meeting, Virtual, November 16-20, 2020. AIChE abstract book. 2020 American Institute of Chemical Engineers Annual Meeting, Virtual, November 16-20, 2020. , Nov-2020
Abstracts for Journals and Proceedings	Brosius N, Ward K, Wilson E, Karpinsky Z, SanSoucie M, Ishikawa T, Matsumoto S, Narayanan R. "Surface Tension Measurement Using Resonance of Levitated Molten Droplets." Presented at the 73rd Annual Meeting of the American Physical Society (APS) Division of Fluid Dynamics, Virtual, November 22-24, 2020. APS Division of Fluid Dynamics List of Abstracts, November 2020. , Nov-2020
Abstracts for Journals and Proceedings	Brosius N, Ward K, Wilson E, Karpinsky Z, SanSoucie M, Ishikawa T, Matsumoto S, Narayanan R. "Surface Tension Measurement Using Resonance of Levitated Molten Droplets." Presented at the 2020 International Space Station Research and Development Conference (ISSRDC), Virtual Conference, July-October, 2020. ISS, R&D Abstracts Online. 2020 International Space Station Research and Development Conference (ISSRDC), Virtual Conference, July-October, 2020. , Jul-2020
Articles in Peer-reviewed Journals	Brosius N, Ward K, Wilson E, Karpinsky Z, SanSoucie M, Ishikawa T, Matsumoto S, Narayanan R. "Benchmarking surface tension measurement method using two oscillation modes in levitated liquid metals." npj Microgravity. 2021 Mar 9;7(1):10. https://doi.org/10.1038/s41526-021-00137-9 ; PMID: 33750800; PMCID: PMC7943785 , Mar-2021
Articles in Peer-reviewed Journals	Dinesh B, Narayanan R. "Branching behaviour of the Rayleigh–Taylor instability in linear viscoelastic fluids." Journal of Fluid Mechanics. 2021 May 25;915:A63. https://doi.org/10.1017/jfm.2021.80 , May-2021
Patents	U.S. Patent No. 10641694. Issue Date. May 5 2020. May-2020 Ward KL, Narayanan R. "Measuring Interfacial Tension Using Electrostatic Oscillations."